

Homework 2 - Problem 1

CS 6347

Christian Parker
CXP220034

March 16, 2025

1. The smallest $2n$ -cycle with $n \geq 2$ is the 4-cycle graph. When $0 < k < 2$, every vertex has only one color option. Since every vertex in the 4-cycle graph has two neighbors and k -coloring requires neighboring vertices to be different colors, the graph cannot be colored.
2. Given the new potential function, the MAP assignment would be:

$$\hat{X} = \arg \max_X \frac{1}{Z} \prod_{i \in V} e^{w_{x_i}} \prod_{(i,j) \in E} 1_{x_i \neq x_j}$$

To determine the maximum estimate, we only need to find the valid assignment that maximizes the potential function $\phi_i(x_i) = e^{w_{x_i}}$ across all vertices. Z is the same for all color assignments and the indicator function simply determines if the coloring is valid and doesn't contribute the probability beyond that. Therefore, the estimate is maximized when $\prod_{i \in V} e^{w_{x_i}}$ is maximized. This function increases exponentially with w_{x_i} and $w_a = a$. Therefore, we can simplify the problem to the following:

$$\hat{X} = \arg \max_X \prod_{i \in V} x_i \prod_{(i,j) \in E} 1_{x_i \neq x_j}$$

This MAP assignment will assign $x_i = k$ to as many vertices as possible, followed by $k - 1$, and so on. The fewest colors required for a $2n$ -cycle graph is two as each vertex will be assigned one color while both of its neighbors are assigned the other. So, the MAP assignment for any $2n$ -cycle graph would be the graph such that every vertex alternates between k and $k - 1$.

3. The marginal probability distribution over any variable x_i is given by:

$$p(x_i) = \sum_{X \setminus x_i} \left(\frac{1}{Z} \prod_{i \in V} e^{x_i} \prod_{(i,j) \in E} 1_{x_i \neq x_j} \right)$$

For example, if $x_1 = 1$, then the valid colorings would be:

x_1	x_2	x_3	x_4	$\prod_{i \in V} e^{x_i}$
1	2	1	2	e^6
1	2	1	3	e^7
1	2	3	1	e^7
1	2	3	2	e^8
1	3	1	2	e^7
1	3	1	3	e^8
1	3	2	1	e^7
1	3	2	3	e^9

Therefore:

$$\begin{aligned}
p(x_1 = 1) &= \sum_{X \setminus x_1} \left(\frac{1}{Z} \prod_{i \in V} e^{x_i} \prod_{(i,j) \in E} 1_{x_i \neq x_j} \right) \\
&= \frac{1}{Z} (e^6 + e^7 + e^7 + e^8 + e^7 + e^8 + e^7 + e^9) \\
&= \frac{1}{Z} (e^6 + 4e^7 + 2e^8 + e^9)
\end{aligned} \tag{1}$$

Repeating this for all possible values of x_i will give the marginal probability distribution over x_i .